Srinivasan Raghunathan
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Education:

- Doctor of Philosophy in Astronomy, Universidad de Chile, Chile: Mar 2010 - Feb 2016.
  - **Awarded graduate student of 2013 prize, Department of Astronomy, U. de Chile.**
  - **Awarded CONICYT PhD fellowship by the Chilean National Commission for Scientific and Technological Research.**
- MSc by research in Astrophysics, University of Central Lancashire, UK: Oct 2008 - Jan 2010.
- Bachelor of Engineering, Electronics and Communication, Anna University, India: Aug 2002 - Jun 2006.
  - **85% + distinction.**
  - **Awarded class topper of 2006 prize.**

Appointments:

- University of California Davis, USA (*Current*): Assistant Project Scientist.
- National Centre for Supercomputing Applications, University of Illinois at Urbana-Champaign, USA (2021-2025): Survey Science Fellow at the Centre for AstroPhysical Surveys.
- University of California, Los Angeles, USA (2018 - 2021): Postdoctoral scholar.
- University of Melbourne, Australia (2015 - 2018): Postdoctoral scholar.
- Universidad de Chile, Chile (2011 - 2014): Masters and undergraduate co-lecturer: Cosmology and general astronomy (4 hours per week).
- Electronic Data Systems (EDS), India (2006 - 2008): Software Engineer: Technology - IBM mainframes and UNIX.

Grants / Academic Credentials:

- 2025: Fermi National Accelerator Laboratory - Universities Research Association's Visiting Scholars Program (VSP): 20,000 USD towards cosmological inference from SPT-3G data.
- 2021: Survey Science Fellowship, Centre for AstroPhysical Surveys, National Centre for Supercomputing Applications.
- 2017: Laby-Betty travel grant, University of Melbourne.
- 2015: Simons foundation grant, Aspen centre for Physics.
- 2013: Research student of the year, Department of Astronomy, UChile.
- 2013: Universidad de Chile scholarship to carry out PhD thesis work at Princeton University for three months.
- 2010 - 2015: CONICYT PhD fellowship, National Commission of Scientific Research and Technology, Chile.
- 2006: Graduate student of the year prize, Sri Sairam Engineering College, India.

Student supervisions:

- Achintya Krishnan, PhD (Ongoing). **Topic**: Kinematic Sunyaev–Zeldovich effect and the large-scale structures of the Universe. Co-supervision with Prof. Joaquin Vieira, **University of Illinois, Urbana Champaign**, USA.
- Ana Carolina Oliveira, PhD (Ongoing). **Topic**: Thermal Sunyaev–Zeldovich effect and gravitational lensing. Co-supervision with Prof. Kimmy Wu, **Caltech/Stanford University**, USA.
- Kevin Levy, PhD (Ongoing). **Topic**: CMB lensing and the large-scale structures of the Universe. Co-supervision with Prof. Christian Reichardt, **University of Melbourne**, Australia.
- Karthik Prabhu, PhD (2025). **Topic**: Primary CMB and gravitational lensing. Co-supervision with Prof. Lloyd Knox, **University of California, Davis**, USA. **No. of publications: 2**.
- Eduardo Schiappucci, PhD (2024). **Topic**: Kinematic Sunyaev–Zeldovich effect. Co-supervision with Prof. Christian Reichardt, **University of Melbourne**, Australia. **No. of publications: 1**.
- Kevin Levy, MSc (2022). **Topic**: CMB lensing. Co-supervision with Dr. Kaustuv Basu, **University of Bonn**, Germany. **No. of publications: 1**.
- Dr. Sanjay Patil, PhD (2021). **Topic**: CMB lensing. Co-supervision with Dr. Christian Reichardt, **University of Melbourne**, Australia. **No. of publications: 4**.
- Dr. Tracey Friday, PhD (2021). **Topic**: Large-scale structures and SNe cosmology. Co-supervision with Dr. Roger Clowes, **University of Central Lancashire**, UK. **No. of publications: 2**.
- Alexia Lopez, MSc (2021). **Topic**: Large-scale structures of the Universe. Co-supervision with Dr. Roger Clowes, **University of Central Lancashire**, UK. **No. of publications: 1**.

Services to Astronomy Community and Professional activities:

- **Telescope Allocation Committee:**
  - Member of the National Radio Astronomy Observatory (NRAO) Science Review Panel (Nov 2021 - Nov 2023).
- **Collaboration responsibilities and leadership roles:**
  - Co-ordinator of AtLAST's primordial universe working group (WG, Apr 2025 - to date).
  - Co-ordinator of SPT CMB secondaries and cross-correlations WG (Oct 2024 - to date).
  - SPT x Euclid Project management committee (September 2024 - to date).
  - Co-ordinator of CMB-S4 galaxy clusters WG (May 2021 - to date).
  - Co-ordinator of SPT data analysis WG calls (Jan 2021 - to date).
- **Reviewer:**
  - **Journals:**
    * AstroPhysics Journal (Sep 2025 - to date).
    * Physical Review Letters (May 2021 - to date).
    * Physical Review D. (Sep 2020 - to date).
    * Journal of Cosmology and Astroparticle Physics (Feb 2024 - to date).
    * Monthly Notices of the Royal Astronomical Society (May 2024 - to date).
    * Galaxies, Multidisciplinary Digital Publishing Institute (MDPI) (Sep 2021 - to date).

- NSF 2025 Graduate Research Fellowship Program (GRFP) (Nov 2024 - to date).

- **Tutorials:**
  - Tutor at the computing boot camp (August 2023) - Organised by Centre for AstroPhysics Surveys, UIUC.

- **Colloquium / Seminar organisation:**
  - Astrophysics/Cosmology seminar organiser, Department of Physics and Astronomy, UC Davis (Jan 2026 -).
  - Colloquium organiser, Centre for AstroPhysics Surveys, UIUC (Jan 2022 - Aug 2023).
    - Developed and actively maintaining https://caps.ncsa.illinois.edu/home/talks/.
  - Astrophysics colloquium organiser, School of Physics, U. of Melbourne (Feb 2017 - April 2018).
  - Cosmology journal organiser, School of Physics, U. of Melbourne (Feb 2016 - Feb 2017).
  - Journal club organiser, Department of Astronomy, U. of Chile (Feb 2012 - Feb 2013).

Collaborations:

Present: South Pole Telescope (SPT), CMB-S4, South Pole Observatory (BICEP+SPT), LSST-DESC, Terahertz Intensity Mapper (TIM), CMB-HD and Atacama Large Aperture Submillimeter Telescope (AtLAST).
Past: Atacama B-mode Search (PhD thesis project).

Publications

As top-tier author: arXiv repo. (Papers led by Students / Junior Postdocs are marked with [✯] and also highlighted in green.)

1. **S. Raghunathan**, A. Mitra, N. Šarčević, et al., LSST-DESC Collaboration, "Probing Physics Beyond the Standard Model through Combined Analyses of Next-Generation Type Ia Supernova, CMB, and BAO Surveys", accepted for publication in ApJ, **2026**, arXiv: 2603.09973.
2. **S. Raghunathan**, P. Ade, D. Anbajagane, et al., SPT and Herschel Collaboration, "Measurement of the Full Shape of the Thermal Sunyaev-Zeldovich Power Spectrum from South Pole Telescope and *Herschel*-SPIRE Observations", ApJ Letters, **1002**, L22, **2026**, arXiv: 2602.10107. (Selected to deliver a talk for the "AAS Journal Author Series", hosted by ApJ Letters.)
3. A. Maniyar, F. Bianchini, K. Wu, **S. Raghunathan**, et al., 'SPT-3G D1: Compton-y maps using data from the SPT-3G and Planck surveys", submitted to **PRD**, **2026**, arXiv: 2602.11279.
4. [✯] K. Prabhu, **S. Raghunathan**, E. Anderes et al., "Learning Correlated Astrophysical Foregrounds with Denoising Diffusion Probabilistic Models", JCAP, **2025**, 25, **2025**, arXiv: 2506.09036.
5. [✯] M. Doohan, M. Millea, **S. Raghunathan**, et al., "Quantifying Bias due to non-Gaussian Foregrounds in an Optimal Reconstruction of CMB Lensing and Temperature Power Spectra", JCAP, **2025**, 23, **2025**, arXiv: 2502.20801.
6. [✯] E. Schiappucci, **S. Raghunathan**, C. To et al., "Constraining cosmological parameters using the pairwise kinematic Sunyaev-Zel'dovich effect with CMB-S4 and future galaxy cluster surveys", PRD, **111**, 6, **2024**, arXiv: 2409.18368.
7. [✯] B. Ansarinejad, **S. Raghunathan**, T. M. C. Abbott et al., "Mass calibration of DES Year-3 clusters via SPT-3G CMB cluster lensing", JCAP, **07**, 024, **2024**, arXiv: 2404.02153.

8. [✯] K. Prabhu, **S. Raghunathan**, M. Millea et al., SPT Collaboration, "Testing the ΛCDM Cosmological Model with Forthcoming Measurements of the Cosmic Microwave Background with SPT-3G", ApJ, **973**, 1, **2024**, arXiv: 2403.17925.

9. **S. Raghunathan**, P. Ade, A. Anderson et al., SPT and Herschel Collaboration, "First Constraints on the Epoch of Reionization Using the non-Gaussianity of the Kinematic Sunyaev-Zel'dovich Effect from the South Pole Telescope and *Herschel*-SPIRE Observations", PRL, **1133**, 12, **2024**, arXiv: 2403.02337.

10. L. Di Mascolo, Y. Perrott, ... **S. Raghunathan** et al., AtLAST Collaboration, "Atacama Large Aperture Submillimeter Telescope (AtLAST) Science: Resolving the Hot and Ionized Universe through the Sunyaev-Zeldovich effect", submitted to Open Research Europe, **2024**, arXiv: 2403.00909.

11. D. Jain, T. Choudhury, **S. Raghunathan** et al., "Probing the Physics of Reionization Using kSZ Power Spectrum from Current and Upcoming CMB Surveys", MNRAS, **530**, 1, **2023**, arXiv: 2311.00315.

12. [✯] K. Levy, **S. Raghunathan**, K.Basu, "A Foreground-Immune CMB-Cluster Lensing Estimator", JCAP, **08**, 020, **(2023)**, arXiv: 2305.06326.

13. **S. Raghunathan**, Y. Omori, "A Cross-Internal Linear Combination Approach to Probe the Secondary CMB Anisotropies: Kinematic Sunyaev-Zel'dovich Effect and CMB Lensing", ApJ, **954**, 17, (**2023**), arXiv: 2304.09166.

14. **S. Raghunathan**, "Assessing the Importance of Noise from Thermal Sunyaev-Zel'dovich Signals for CMB Cluster Surveys and Cluster Cosmology", ApJ, **928**, 16 **(2022)**, arXiv: 2112.07656.

15. **S. Raghunathan**, N. Whitehorn, M. Alvarez et al., "Constraining Cluster Virialization Mechanism and Cosmology using Thermal-SZ-selected clusters from Future CMB Surveys", ApJ, **926**, 172 **(2022)**, arXiv: 2107.10250.

16. **S. Raghunathan**, S. Nadathur, B. Sherwin et al., "The Gravitational Lensing Signatures of BOSS Voids in the Cosmic Microwave Background", ApJ, **890**, 168 **(2020)**, arXiv: 1911.08475.

17. **S. Raghunathan**, S. Patil, E. Baxter et al., "A Detection of CMB-Cluster Lensing using Polarization Data from SPTpol", Phys. Rev. Letters, **123**, 181301 **(2019)**, arXiv: 1907.08605. (Selected as editor's suggestion by Phys. Rev. Letters.)

18. [✯] S. Patil, **S. Raghunathan**, C. Reichardt, "Suppressing the thermal SZ-induced variance in CMB-cluster lensing estimators", ApJ, **888**, 9 **(2020)**, arXiv: 1905.07943.

19. **S. Raghunathan**, G. Holder, J. Bartlett et al., "An Inpainting Approach to Tackle the Kinematic and Thermal SZ Induced Biases in CMB-Cluster Lensing Estimators", JCAP, **11**, 037 **(2019)**, arXiv: 1904.13392.

20. E. Baxter, B. Sherwin, **S. Raghunathan**, "Constraining the Rotational Kinematic Sunyaev-Zel'dovich Effect in Massive Galaxy Clusters", JCAP, **6**, 001 **(2019)**, arXiv: 1904.04199.

21. K. Aylor, M. Joy, ... **S. Raghunathan** et al., "Sounds Discordant: Classical Distance Ladder & ΛCDM -based Determinations of the Cosmological Sound Horizon', ApJ, **874**, 4 **(2019)**, arXiv: 1811.00537.

22. **S. Raghunathan**, S. Patil, E. Baxter et al., "Mass Calibration of Optically Selected DES clusters using a Measurement of CMB-Cluster Lensing with SPTpol Data", ApJ, **872**, 170 **(2019)**, arXiv: 1810.10998.

23. **S. Raghunathan**, F. Bianchini, C. Reichardt, "Imprints of gravitational lensing in the Planck CMB data at the location of WISExSCOS galaxies", Phys. Rev. D, **98**, 4 **(2018)**, arXiv: 1710.09770.

24. [✬] T. Friday, R. Clowes, **S. Raghunathan** et al., “Accidental deep field selection bias in CMB temperature and SNe redshift correlation”, MNRAS, **479**, 1137 **(2018)**, arXiv: 1805.09581.

25. A. Kusaka, T. Essinger-Hileman, ... **S. Raghunathan** et al., “Results from the Atacama B-mode Search (ABS) Experiment”, JCAP, **09** 005 **(2018)**, arXiv: 1801.01218. (PhD thesis project.)

26. E. Baxter, **S. Raghunathan**, T. Crawford et al., “A Measurement of CMB Cluster Lensing with SPT and DES Year 1 Data”, MNRAS, **476**, 2674 **(2018)**, arXiv: 1708.01360.

27. **S. Raghunathan**, S. Patil, E. Baxter et al., “Measuring galaxy cluster masses with CMB lensing using a Maximum Likelihood estimator: Statistical and systematic error budgets for future experiments”, JCAP, **08**, 030 **(2017)**, arXiv: 1705.00411.

28. **S. Raghunathan**, R. Clowes, L. Campusano et al., “Intervening Mg II absorption systems from the SDSS DR12 quasar spectra”, MNRAS, **463**, 2640 **(2016)**, arXiv: 1608.05112.

29. R. Clowes, L. Haberzettl, **S. Raghunathan** et al., “Ultraviolet Fe II Emission in Fainter Quasars: Luminosity Dependences, and the Influence of Environments”, MNRAS, **460**, 1428 **(2016)**, arXiv: 1604.08411.

30. S.Simon, **S. Raghunathan**, J. Appel, et al., “Characterization of the Atacama B-mode Search”, SPIE, **9153**, 15 **(2014)**.

31. R. Clowes, **S. Raghunathan**, I. Soechting et al., “Environments of strong / ultrastrong, ultraviolet Fe II emitting quasars”, MNRAS, 433, 2467 **(2013)**, arXiv: 1304.7396.

32. R. Clowes, K. Harris, **S. Raghunathan** et al., “A structure in the early Universe at z $\sim$1.3 that exceeds the homogeneity scale of the R-W concordance cosmology”, MNRAS, 429, 2910 **(2013)**, arXiv: 1211.6256.

As contributing author:

1. P. Chaubal, N. Huang, ... **S. Raghunathan** et al., “SPT-3G D1: A Measurement of Secondary Cosmic Microwave Background Anisotropy Power”, submitted to JCAP, **2026**, arXiv: 2601.20551.

2. A. Mitra, R. Kessler, ... **S. Raghunathan** et al., “A Fully Photometric Approach to Type Ia Supernova Cosmology in the LSST Era: Host Galaxy Redshifts and Supernova Classification”, submitted to PRD, **2025**, arXiv: 2512.06319.

3. E. Camphuis, W. Quan, ... **S. Raghunathan** et al., “SPT-3G D1: CMB temperature and polarization power spectra and cosmology from 2019 and 2020 observations of the SPT-3G Main field”, Accepted by PRD, **2025**, arXiv: 2506.20707.

4. M. Archipley, A. Hryciuk, ... **S. Raghunathan** et al., “Millimeter-wave observations of Euclid Deep Field South using the South Pole Telescope: A data release of temperature maps and catalogs’, A&A, **706**, A17, **2025**, arXiv: 2506.00298.

5. C. Papovich, J. Cole, ... **S. Raghunathan** et al., “Galaxies in the Epoch of Reionization Are All Bark and No Bite — Plenty of Ionizing Photons, Low Escape Fractions’, Accepted to ApJ, **2025**, arXiv: 2505.08870.

6. J. Zebrowski, C. Reichardt, ... **S. Raghunathan** et al., “Constraints on Inflationary Gravitational Waves with Two Years of SPT-3G Data’, PRD, **112**, 12**2025**, arXiv: 2505.02827.

7. F. Qu, F. Ge, ... **S. Raghunathan** et al., “Unified and consistent structure growth measurements from joint ACT, SPT and *Planck* CMB lensing’, submitted to PRL, **2025**, arXiv: 2504.20038.

8. E. Hughes, F. Ge, ... **S. Raghunathan** et al., “A cool dark sector, concordance, and a low σ_8”, PRD, **6**, **2023**, arXiv: 2311.05678.

9. J. Hernandez, L. Bleem, ... **S. Raghunathan** et al.“Dissecting the Thermal SZ Power Spectrum by Halo Mass and Redshift in SPT-SZ Data and Simulations”, OjA, **6**, 41, **2023**, arXiv: 2309.12475.
10. Z. Pan, F. Bianchini, ... **S. Raghunathan** et al.“A Measurement of Gravitational Lensing of the Cosmic Microwave Background Using SPT-3G 2018 Data”, PRD, **108**, 12, **2023**, arXiv: 2308.11608.
11. A. Gardener, E. Baxter, **S. Raghunathan** et al.“Prospects for studying the mass and gas in protoclusters with future CMB observations”, OjA, **7**, 2, **2023**, arXiv: 2307.15309.
12. J. Sanchez, Y. Omori, ... **S. Raghunathan** et al., “Mapping gas around massive galaxies: cross-correlation of DES Y3 galaxies and Compton-y-maps from SPT and *Planck*”, MNRAS, **2**, 3163 **(2022)**, arXiv: 2210.08633.
13. A. Anderson, P. Barry, ... **S. Raghunathan** et al., “SPT-3G+: Mapping the High-Frequency Cosmic Microwave Background Using Kinetic Inductance Detectors”, SPIE, **12190**, 1219003, **(2022)**, arXiv: 2208.08559.
14. K. Dibert, A. Anderson, ... **S. Raghunathan** et al., “Forecasting ground-based sensitivity to the Rayleigh scattering of the CMB in the presence of astrophysical foregrounds”, accepted in Phys. Rev. D, **106**, 6 **(2022)**, arXiv: 2205.04494.
15. The CMB-S4 Collaboration, “Snowmass 2021 CMB-S4 White Paper”, arXiv: 2203.08024.
16. The CMB-HD Collaboration, “Snowmass2021 CMB-HD White Paper”, arXiv: 2203.05728.
17. L. Bleem, T. Crawford, ... **S. Raghunathan** et al., “CMB/kSZ and Compton-y Maps from 2500 square degrees of SPT-SZ and Planck Survey Data”, ApJ, **258**, 2 **(2022)**, arXiv: 2102.05033.
18. D. Dutcher, L. Balkenhol, ... **S. Raghunathan** et al., “Measurements of the E-Mode Polarization and Temperature-E-Mode Correlation of the CMB from SPT-3G 2018 Data”, Phys. Rev. D **104**, 2 **(2021)**, arXiv: 2101.01684.
19. CMB-S4 collaboration, “CMB-S4 Science Case, Reference Design, and Project Plan”, **(2019)**, arXiv: 1907.04473.
20. J. Avva, P. Ade, ... **S. Raghunathan** et al., “Particle Physics with the Cosmic Microwave Background with SPT-3G’, **(2019)**, arXiv: 1911.08047.
21. K. Basu, M. Remazeilles, ... **S. Raghunathan** et al., “A Space Mission to Map the Entire Observable Universe using the CMB as a Backlight”, **(2019)**, arXiv: 1909.01592.
22. J. Delabrouille, M. Abitbol, ... **S. Raghunathan** et al., “Microwave Spectro-Polarimetry of Matter and Radiation across Space and Time”, **(2019)**, arXiv: 1909.01591.
23. A. Bender, P. A. R. Ade, ... **S. Raghunathan** et al., “Year two instrument status of the SPT-3G cosmic microwave background receiver”, SPIE, **10708**, 1070803 **(2018)**, arXiv: 1809.00036.
24. T. Essinger-Hileman, A. Kusaka, ... **S. Raghunathan** et al., “Systematic effects from an ambient-temperature, continuously-rotating half-wave plate”, Review of Scientific Instruments, **87 (2016)**, arXiv: 1601.05901.
25. S. Simon, J. Appel, ... **S. Raghunathan** et al., “Characterization of the Atacama B-mode Search Detectors with a Half-Wave Plate”, Journal of Low Temperature Physics, **184**, 534 **(2016)**, arXiv: 1511.04760.
26. A. Kusaka, T. Essinger-Hileman, ... **S. Raghunathan** et al., “Modulation of CMB polarization with a warm rapidly rotating half-wave plate on the ABS instrument”, Review of Scientific Instruments, **85 (2014)**, arXiv: 1310.3711.

27. S. Simon, J. Appel, ... **S. Raghunathan** et al., "In Situ Time Constant and Optical Efficiency Measurements of TRUCE Pixels in the Atacama B-Mode Search", Journal of Low Temperature Physics, **176**, 712 **(2014)**.

Conference/meeting talks: ✯- Invited/Review; Black - Conference; Red - Colloquium / Seminar.

- ✯ "Cosmology using Stage-IV Surveys", The Concurrence of Mega-Surveys (Unravelling a multi-messenger view of our Universe using a synergy of advanced analysis techniques, machine learning and extremely large datasets), International Centre for Theoretical Sciences (ICTS), India, Aug, 2026.
- ✯ "Sunyaev-Zeldovich Science and Challenges", Exploring New Boundaries in Cosmology and Astrophysics: Cosmic Microwave Background and Large Scale Structure Surveys, Kavli Institute for Theoretical Physics (KITP), Santa Barbara, Mar, 2026.
- " Cosmic Microwave Background Science Using the South Pole Telescope and Beyond", Stellenbosch University, Mar 2026.
- " Cosmic Microwave Background Science Using the South Pole Telescope and Beyond", Beus Seminar Series, Arizona State University, Dec 2025.
- "Measurement of the small-scale thermal Sunyaev–Zeldovich power spectrum with the South Pole Telescope", Texas Symposium on Relativistic Astrophysics, Arizona, Dec 2025.
- ✯ "South Pole Telescope", CMB-Bharat Seminar Series, Dec 2024.
- ✯ "South Pole Telescope x Euclid", Euclid CMBXC WG Meeting, Kavli Institute for Cosmology, University of Cambridge, Nov 2024.
- "Cosmology and Astrophysics with the Secondary Anisotropies of the Cosmic Microwave Background from South Pole Telescope and Future Surveys.", University of New Mexico, New Mexico State University, Texas A&M University, University of California Irvine, University of California Los Angeles, California Institute of Technology, University of Texas at Austin, USA, Oct-Dec 2024.
- ✯ "Review talk on the kinematic Sunyaev–Zeldovich Effect", New Physics from Old Light: Illuminating the Universe with CMB Secondaries, University of Cambridge, September 2024..
- ✯ "South Pole Telescope", CMB-S4 meeting, University of Illinois Urbana Champaign, July 2024.
- ✯ "Constraining the Epoch of Reionisation using CMB as the backlight", IoA50: New Frontiers of Astronomy, University of Cambridge, July 2024.
- ✯ "Constraints on the Epoch of Reionisation using the kSZ 4-pt measurement from the South Pole Telescope", PASCOS 2024 - Rencontres du Vietnam, July 2024.
- ✯ "Cosmology with the South Pole Telescope (SPT): Latest and Upcoming results from SPT", Cosmology from home, June 2024.
- "Sunyaev-Zeldovich from CMB Surveys", University of Southern California / University of California Santa Cruz, USA, March 2024.
- "Sunyaev-Zeldovich from CMB Surveys", University of Cincinnati / Case Western Reverse University, USA, Nov 2023.
- "Constraining the Epoch of Reionisation Using the kinematic Sunyaev-Zeldovich Signal.", University of Illinois Urbana Champaign / University of Minnesota, USA, Sep/Oct 2023.

- "Prospects for Kinematic Sunyaev-Zeldovich Measurements from Current and Future CMB Experiments", Observing the Universe at millimetre wavelengths, Grenoble, France, June 2023.
- "Towards a robust detection of the kinematic Sunyaev-Zeldovich power spectrum using South Pole Telescope and Herschel-SPIRE data.", California Institute of Technology, University of California Davis, and University of Wisconsin Milwaukee, USA, April/May 2023.
- ✯ "Observing and interpreting the most ancient light in the universe", Mini School on Gravitation and Cosmology, Indian Institute of Technology IIT-Madras, India, Nov 2022. (Invited Review Talk)
- "Demystifying the dark side of the universe Using kinematic and thermal Sunyaev-Zeldovich effects.", Tata Institute of Fundamental Research, Nov 2022.
- ✯ "Kinematic Sunyaev–Zeldovich effect and reionisation science from small-scale CMB experiments", CMB+EoR Summer workshop, Montreal, Canada, July 2022. (Invited Review Talk)
- "First detection of polarized CMB-Cluster lensing using SPTpol and DES", Cluster Mass ESA meeting, Virtual, Sep 2021.
- "Demystifying the dark side of the universe using secondary cosmic microwave background anisotropies.", USC, USA, Jan 2021 and NCSA, USA, May 2021.
- "CMB-Cluster lensing", Stanford / UCLA, USA, Jan 2018; and Harvard / Cornell / Princeton, USA, Sep 2017.
- "CMB-Cluster lensing forecasts for the CMB-S4 experiment", Clusters 2017, Spain, Jul 2017.
- "CMB-Cluster lensing with SPTpol and SPT-3G: Forecasts and systematic error budgets", Swinburne University and University of Queensland, Australia, Jun 2017.
- "Primordial gravitational waves and CMB polarization", University of Melbourne, School of Physics colloquium, Australia, Aug 2016.
- "Towards the first detection of CMBPol cluster lensing using SPTpol data", Australian Astronomical Society Annual meeting, University of Sydney, Australia, Jul 2016.
- "Status of the Atacama B-mode Search experiment", Cosmology on the Beach, Mexico, Jan 2016.
- "Search for primordial gravitational waves using Atacama B-mode Search experiment", SOCHIAS (Chilean Astronomical Society) annual meeting, Chile, Mar 2015.
- "Search for primordial B-modes from CMB polarization and characterizing ABS telescope", Alpine cosmology workshop, Austria, Jul 2014.
- "Pointing and beam characterization of the Atacama B-mode Search (ABS) experiment", Workshop on New Light in Cosmology from the CMB, ICTP, Italy, Jul 2013.
- "CMB Polarization and the search for tensor modes", CosmoSur II, Chile, May 2013.
